Sanika Khanduja

sanakhanduja@gmail.com | 8287403095 | LeetCode | Codeforces | GitHub | LinkedIn

Education

Indira Gandhi Delhi Technical University for Women (IGDTUW) **Jul 2023 – Jun 2027**
- *Bachelor of Technology – Computer Science & Engineering* *Delhi, India*
  CGPA: 9.72 | Rank 4/231 (CSE)
- **Queen Mary's School Northend** **Jul 2020 – Jun 2023**
  Class XII: 95.8% | Class X: 96.4% *Delhi, India*

Skills

- **Languages: C++**, **Python**, C
- **Core CS: Data Structures & Algorithms**, Object Oriented Programming, Operating Systems, DBMS
- **Backend & Databases:** REST APIs, Node.js, Express.js, **MySQL**, MongoDB
- **Web:** React, **TypeScript**, HTML, CSS, JavaScript
- **AI/ML:** OpenAI, LangChain, **RAG**, Milvus, PyTorch, Scikit-learn, Pandas, NumPy
- **Tools:** Git, GitHub, VS Code, Postman, **Azure**

Experience

Microsoft | Software Engineering Intern **June 2026 – July 2026**
- Built **Disk Layout Impact Detection** (**C#/.NET 8**) for Azure Hardware Provisioning: a deterministic impact-analysis engine that resolves a **3-level inheritance hierarchy** in an **80K+ line** production config to compute the exact blast radius of any change.
- Designed **inverted-index** based reverse lookups for near-**O(1)** subsection → layout → SKU tracing, and a resolved-view **diff engine** covering value, rename, add/delete, and deep-merge dependency changes.
- Shipped the detector across **3 surfaces** (CLI, Copilot skill, PR bot) and an **Azure DevOps Function app** that auto-posts impact comments on every relevant PR in **10–15s**; validated against real production incidents (**59+ machines** impacted) the tool would have caught pre-merge.
- Secured the service with **Managed Identity Authentication** (zero stored secrets), delivering a reusable platform extensible beyond a single config file.

Cisco | Software Engineering Intern **Jun 2025 – Jul 2025**
- Built **CDETS Co-Pilot**, an AI-powered bug-triaging platform using **LangChain** and **RAG** over **30,000+** historical defects, reaching **77%** component-prediction accuracy.
- Built semantic retrieval **REST APIs** with **Azure OpenAI** and **Milvus** across **135+ components** for log analysis and automated fix suggestions, cutting debugging effort.
- Designed the **Bug Filing Guidance (BFG)** module (Azure OpenAI, Ollama) to automate bug-quality validation and one-click PDF reports; in evaluation by **1000+ developers**.

Projects

AutiCareAI (TypeScript, Node.js, Express, Python, OpenCV, Supabase) | **Jul 2026**
- Built an **AI-powered autism care platform** with a **Python/OpenCV** computer-vision engine that analyzes video for behavioral markers (eye contact, facial expressions, social engagement) to support early ASD screening and continuous progress tracking.
- Developed **REST APIs** (**Node.js/Express**) powering a **TypeScript** caregiver dashboard and clinician view, backed by a **Supabase** data layer

Flood Damage Prediction (Python, NumPy, Pandas, Matplotlib, Scikit-learn) | **Aug 2024**
- Built ML regression models on **20+ years** of Indian flood data, achieving $\mathbf{R^2 = 0.98}$ for long-term damage forecasting; co-authored a research paper accepted at the **ICDPN Conference**.

Achievements

- **DSA Proficiency** | Solved **500+ problems** across **LeetCode (rating: 1716)** and **Codeforces (rating: 1154)**
- **IEEE Wiegnite 3.0** | **2nd Runner Up:** Top 3 in the **flagship hackathon of IEEE, IGDTUW**
- **Flipkart Grid 7.0** | **Semi-Finalist:** Top **800 out of 160,000+** in Flipkart's national tech challenge.
- **Flipkart Runway Challenge** | **Top 420:** Advanced to Round 2 among 45,000+ in a national-level contest.
- **Smart India Hackathon** | **Semi-Finalist:** Selected in the top 50 of 1500+ teams.
- **Academic Excellence** | **Rank 1/1000+** in B.Tech Year 1 at IGDTUW (CGPA: **9.91/10**).

Leadership & Responsibilities

- **President, Entrepreneurship Cell (E-Cell), IGDTUW** Leading a **200+ member** team across marketing, operations, and creative to organize hackathons, startup events, and the institute's first flagship **E-Summit** with **1000+ participants**.
- **Competitive Programming Mentor, GDG IGDTUW** Mentored **70+ students** in Data Structures, Algorithms, and Competitive Programming through structured sessions, live coding, and contest preparation.